3.3.26 EXP: NO:- 7

FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS:-

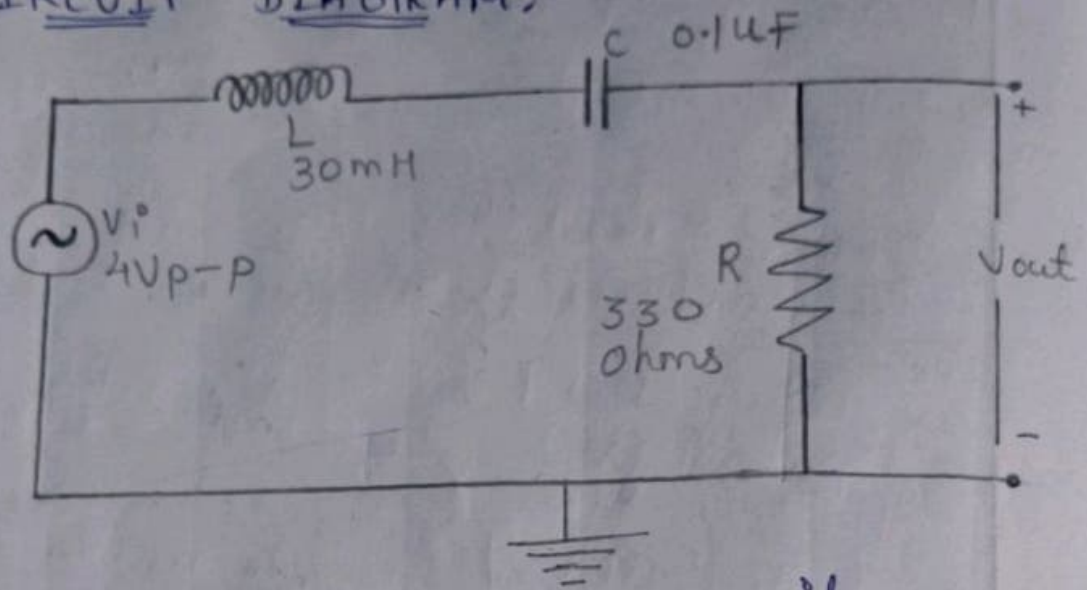
AIM:

To study the behavior of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using Multisim simulator.

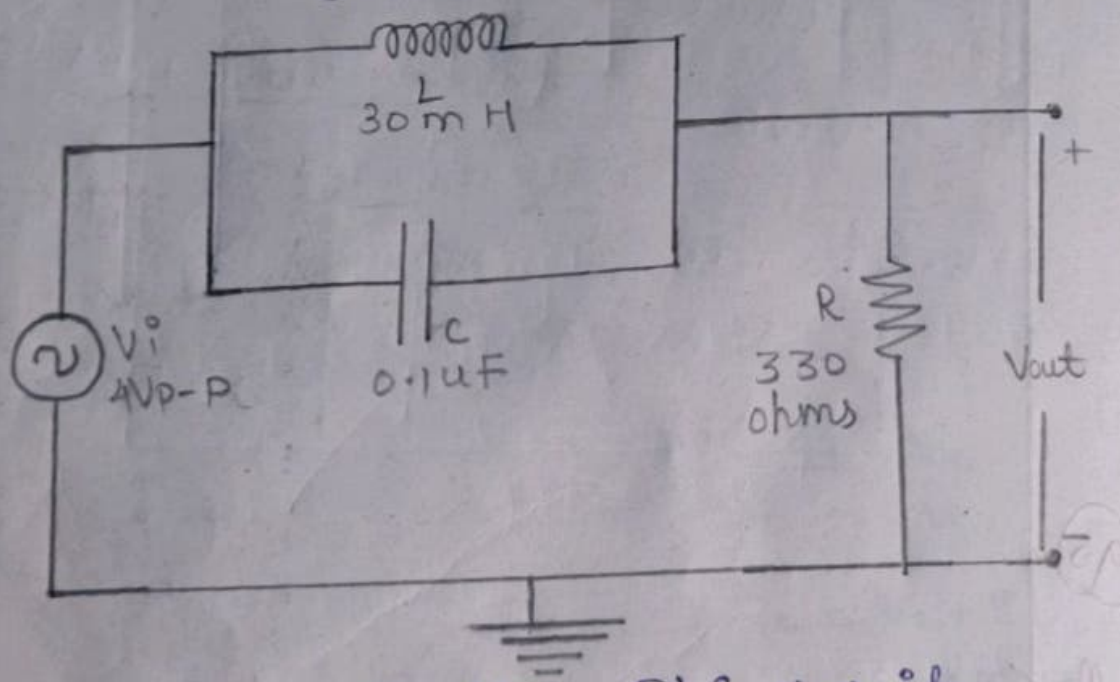
APPARATUS REQUIRED:-

S.no	components	Specification	Quantity
1.	CRO	10 MHz	1
2.	DIB	—	1
3.	Capacitors	0.1 μf	1
4.	Resistors	330Ω	1
5.	Bread board	—	—
6.	Connecting wires	—	—
7.	PC with Multisim simulator	—	—

CIRCUIT DIAGRAM:-



Series RLC circuit



Parallel RLC circuit

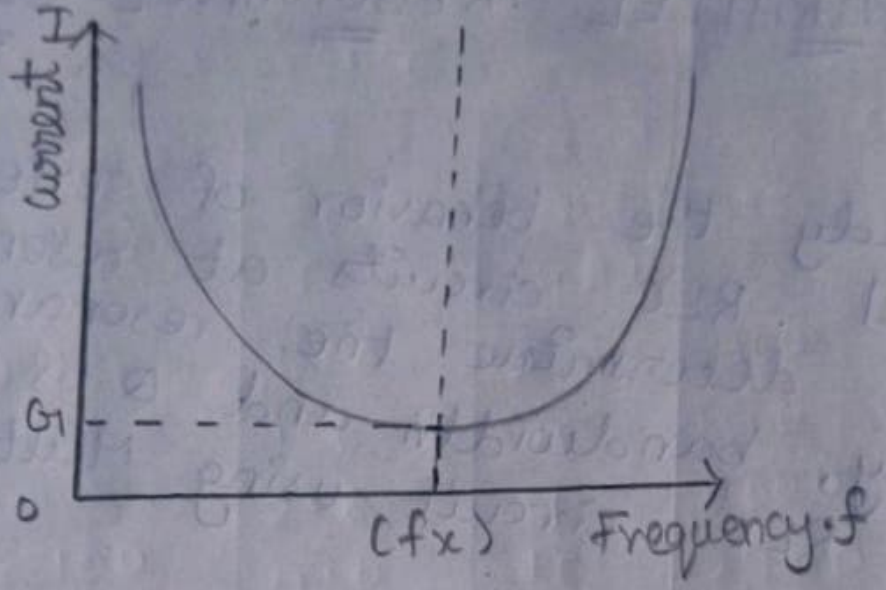
Formulae:

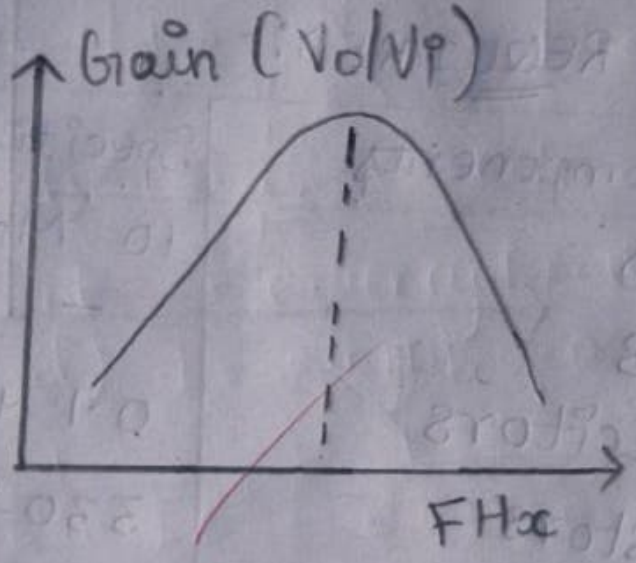
$$f_0 = 1/2\pi\sqrt{LC}$$

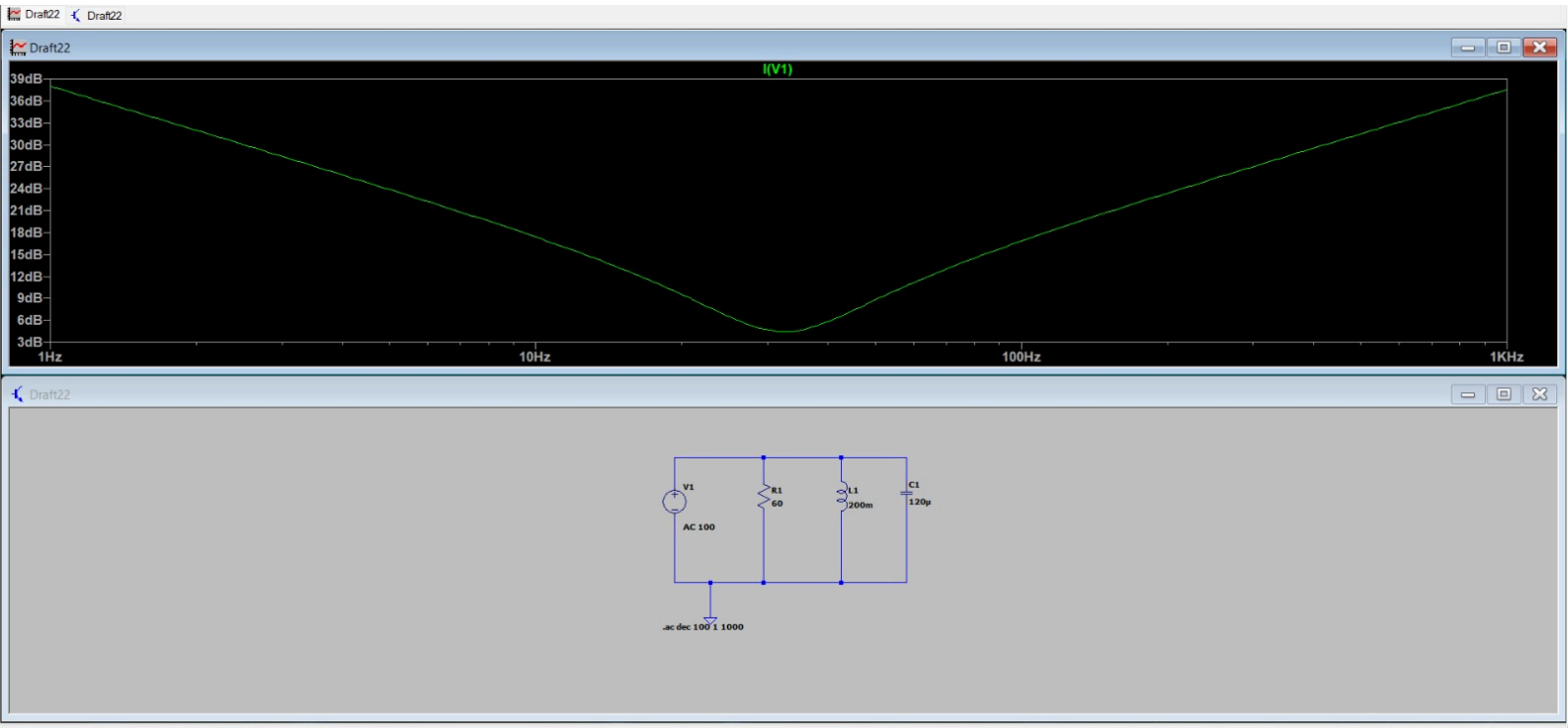
$$BW = f_2 - f_1$$

$$Q = f_0/BW$$

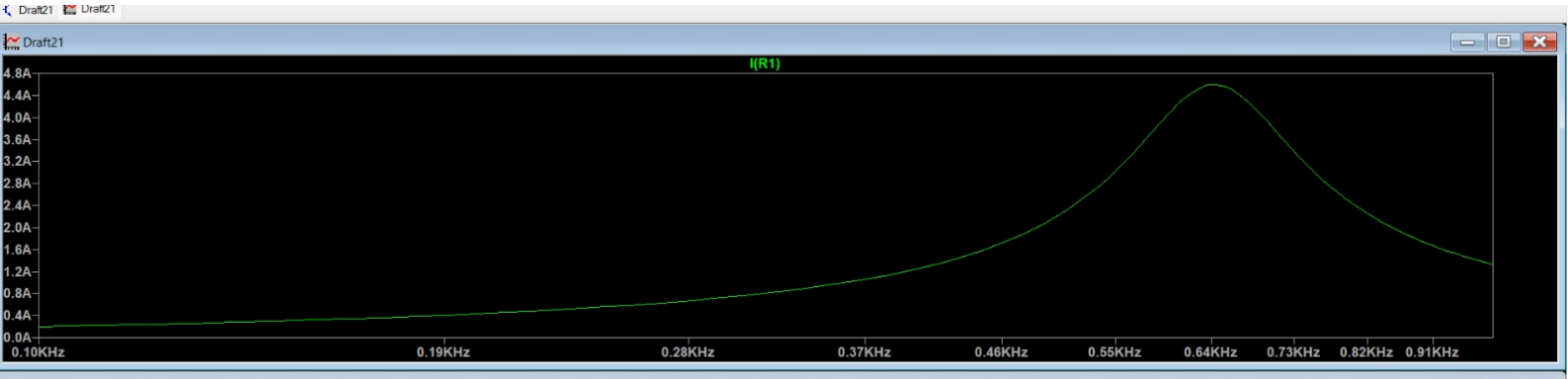
Parallel resonance :-

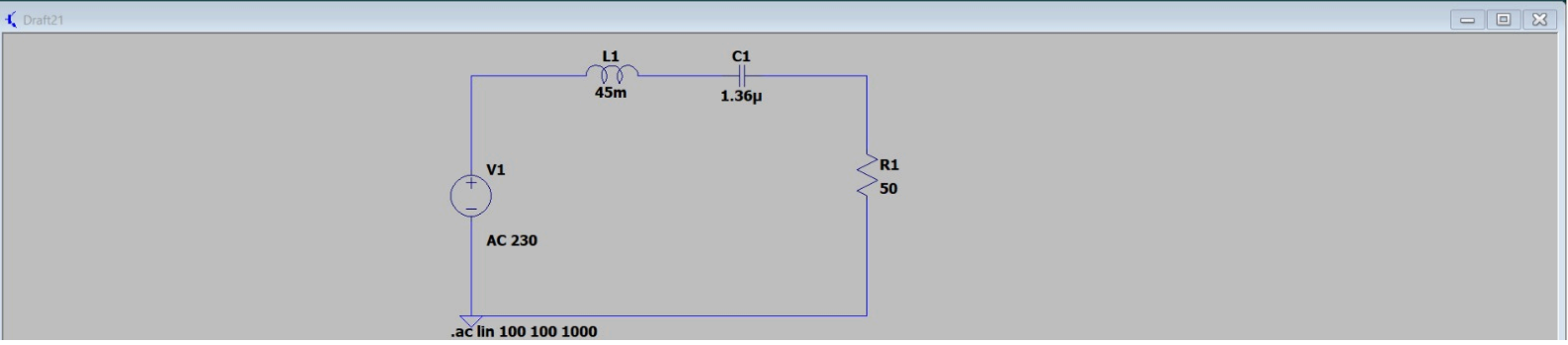






x = 26.718Hz y = 27.708dB


Draft21
I(R1)
4.8A
4.4A
4.0A
3.6A
3.2A
2.8A
2.4A
2.0A
1.6A
1.2A
0.8A
0.4A
0.0A
0.10KHz
0.19KHz
0.28KHz
0.37KHz
0.46KHz
0.55KHz
0.64KHz
0.73KHz
0.82KHz
0.91KHz


Draft21
L1
45m
C1
1.36µ
V1
AC 230
R1
50
.ac lin 100 100 1000

PROCEDURE:-

1. Construct circuit shown in figure 1 on breadboard.
2. Connect CRO ch 1 to input and ch 2 to output.
3. Set the input voltage to 4V p-p.
4. Vary the frequency from 500Hz to 3kHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage V_m is the resonance frequency.
5. If $V_{cutoff} = V_m/\sqrt{2}$, vary the frequency again until the output voltage equals V_{cutoff}. This frequency is the cut-off frequency. There should be 2 cut-off frequencies on either side of resonant frequency.
6. Calculate the bandwidth by subtracting the 2 cut-off frequencies.
7. Calculate the Q factor
8. Repeat steps 1 through 7 for circuit shown in figure 2.

Tabulation: $V_{in} = 4V_{p-p}$

Series Resonance			Parallel Resonance		
Frequency (Hz)	Output voltage (V)	Output current (mA)	Frequency (Hz)	output voltage (V)	Output current (mA)
500	0.8	2	500	3.8	10
1000	1.6	4	1000	3.2	8
2000	2.8	7	2000	2.4	6
2900	4.0	10	2900	2.0	5
3500	2.5	6	3500	2.7	7
4000	1.5	3	4000	3.4	9

Manual calculation :-

Resonant Frequency (Both series and parallel) $L = 30\,mH = 0.03\,H$, $C = 0.1\,\mu F = 1\times10^{-7}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.03\times10^{-7}}} = 2900\,Hz$$

$$f_0 = 2.9\,KHz$$

O/P verified

Assuming : $f_1 = 2.1\,KHz$, $f_2 = 3.7\,KHz$

Bandwidth : $BW = f_2 - f_1$

$= 3700 - 2100 = 1600\,Hz$

Quality Factor (Both series and parallel)

$$Q = \frac{f_0}{BW} = \frac{2900}{1600} = 1.81$$

MARKS SPLIT-UP:

Sl.NO	Description	Marks Allotted	Marks obtained
1.	Experimental Setup/connections	20 marks	20
2.	Observation/ Readings	20 marks	20
3.	Calculations / Results	20 marks	20
4.	Viva	20 marks	20
5.	Record	20 marks	20
		Total	100

Result:

Thus, the phenomenon of resonance in RLC circuit was studied and the following were determined using Multisim simulator.

Series RLC circuit:	Parallel RLC circuit:
Resonant frequency = 2.9 KHZ	Resonant frequency = 2.9 KHZ
Bandwidth = 1600 HZ	Bandwidth = 1600 HZ
Q factor = 1.81	Q factor = 1.81